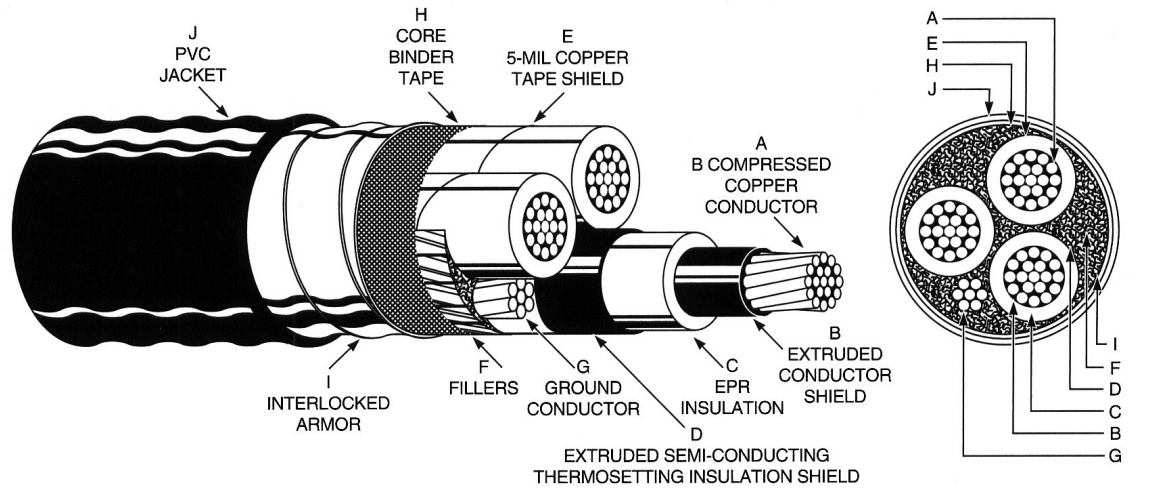
AL09ET

**5kVU or 8kV Type MV-105 or MC. 3 Copper Conductors.
Thermosetting Conductor and Insulation Shield.
EPR Insulation. Tape Shield. Copper Ground Wire.
Aluminum Armor. PVC Jacket.
RoHS**



APPLICATIONS

Southwire AL09ET Type MV-105 or MC Cable is for use in aerial installations, direct burial, concrete-encased installations, cable trays, troughs or continuous rigid cable supports. These cables are capable of operating continuously at a maximum conductor temperature of 105°C for normal operation, 140°C for emergency overload conditions, and 250°C for short circuit conditions, and are rated at 5,000V, 133% (ungrounded system) insulation level, and 8,000V, 100% insulation level (grounded system). Maximum sidewall pressure is 1000 lbs.

SPECIFICATIONS

Southwire AL09ET Type MV-105 or MC Cable is manufactured and tested in accordance with the latest revisions of the following standards and specification:

- UL 1072 - Medium Voltage Power Cables.
- ICEA S-93-639 (NEMA WC 74)
- IEEE 1202/FT4 - Flame Test (70,000 Btu/hr Vertical Tray Test) with CPE and Solonon jacket.
- ICEA T-29-520 - Vertical Cable Tray Flame Test (210,000 Btu/hr).
- AEIC CS-8
- RoHS

CONSTRUCTION

Southwire AL09ET Type MV-105 or MC Cable offers heavy duty construction and flexible EPR insulation. Alternate constructions available. For use in cable trays per NEC Article 392. SOLONON™ low smoke, non halogen polyolefin jackets and CPE jackets are available upon request.

Southwire
One Southwire Drive
Carrollton, Ga. 30119 USA

MADE IN THE USA

greenSpec™ RoHS Compliant

Copyright 2010, Southwire Company. All Rights Reserved.
®Southwire is a registered trademark of Southwire Company.

AL09ET 3c, 5/8kV, 115mil EPR(133/100%), TS, GW, Alum

Product Code	Size	Diameter		0.115" (2.92mm) Insulation Diameter		Extruded Insulation Shield Diameter		Grd. Cond. Size	Armored Diameter		Jacket Thickness		Approx. Overall Diameter		Approx. Net Weight		Allowable Ampacity+	
	AWG/kcmil	inch*	mm	inch*	mm	inch***	mm	AWG	inch	mm	inch	mm	inch	mm	lb./Mft.	kg/km	Direct Burial	In Air
AL09-ET-002	2	.283	7.19	.581	14.76	.623	15.82	6	1.706	43.33	.06	1.52	1.826	46.4	1805	2686	115	145
AL09-ET-001	1	.322	8.18	.621	15.77	.663	16.83	4	1.792	45.52	.06	1.52	1.912	48.6	2081	3096	180	175
AL09-ET-010	1/0	.362	9.20	.661	16.79	.703	17.86	4	1.879	47.73	.06	1.52	1.999	50.8	2376	3535	210	200
AL09-ET-020	2/0	.405	10.29	.701	17.81	.743	18.87	4	1.965	50.10	.06	1.52	2.085	53.0	2724	4053	235	225
AL09-ET-030	3/0	.456	11.58	.751	19.08	.793	20.14	3	2.073	52.65	.06	1.52	2.193	55.7	3189	4745	270	270
AL09-ET-040	4/0	.512	13.00	.806	20.47	.848	21.54	3	2.192	55.68	.06	1.52	2.312	58.7	3714	5526	310	305
AL09-ET-250	250	.558	14.17	.852	21.64	.905	22.99	3	2.316	58.83	.06	1.52	2.466	62.6	4335	6450	345	355
AL09-ET-350	350	.661	16.79	.955	24.26	1.008	25.60	2	2.537	64.44	.06	1.52	2.687	68.2	5477	8149	415	430
AL09-ET-500	500	.790	20.07	1.080	27.43	1.133	28.78	1	2.497	63.42	.06	1.52	2.957	75.1	7228	10755	505	530
AL09-ET-750	750	.968	24.59	1.260	32.00	1.320	33.53	1/0	3.212	81.59	.06	1.52	3.382	86.0	10157	15113	630	665

SCOPE: This specification covers three conductor EPR (ethylene propylene rubber) insulated, shielded, interlock armored, thermoplastic jacketed power cables with grounding conductor for use in aerial installations, direct burial, concrete-encased installations, cable trays, troughs, or continuous rigid cable supports. These cables are capable of operating continuously at a temperature of 105°C for normal operations, 140°C for emergency overload conditions, and 250°C for short circuit conditions, and are rated at 5,000V, 133% insulation level, and 8,000V, 100% insulation level.

STANDARDS: The following standards shall form part of this specification: UL Standard 1072 for Medium Voltage Power Cable and ICEA S-93-639 (NEMA WC 74) 5-46 kV Shielded Power Cable for Use in the Transmission and Distribution of Electric Energy.

CONDUCTOR: The conductor shall be Class B compressed concentric stranded bare copper in accordance with ASTM B3 and B8 and ICEA .

CONDUCTOR SHIELD: The conductor shall be shielded with an extruded semi-conducting thermosetting polymeric layer, which shall be firmly bonded to the insulation. The thickness shall be in accordance with the referenced standards.

INSULATION: The insulation shall be EPR (ethylene propylene rubber) meeting the requirements of the referenced standards. The nominal thickness shall be 0.115".

INSULATION SHIELD: The insulation shall be shielded with an extruded semi-conducting thermosetting polymeric layer which shall be identified as semi-conducting. Over this layer shall be applied a helically-wrapped 5-mil copper tape with 25% overlap. The method of phase identification will be similar to ICEA Method 3, using printed circuit numbers and colors (1-BLACK, 2-RED, 3-BLUE).

GROUNDING CONDUCTOR: The ground conductor shall be Class B compressed concentric stranded bare copper in accordance with ASTM B3 and B8.

ASSEMBLY: The insulated and shielded power conductors shall be cabled round with fillers and with a grounding conductor in one outer interstice and covered with a binder tape.

ARMOR: A single strip of interlocked armor of aluminum shall be applied over the assembly.

JACKET: The cable shall be covered with a yellow PVC jacket conforming to the requirements specified for polyvinyl chloride in ICEA. The average thickness shall be in accordance with the referenced standards. The jacket will be sunlight resistant and will meet the requirements of UL 1685 - UL Flame Exposure Test. Optional SOLONON low smoke, non halogen polyolefin jackets and CPE jackets are available upon request.

IDENTIFICATION: Manufacturer's identification shall be printed on the jacket.

TESTS: Qualification tests shall be conducted in accordance with the requirements of AEIC.

Southwire
One Southwire Drive
Carrollton, Ga. 30119 USA

MADE IN ★ THE USA

greenSpec™
RoHS Compliant

Copyright 2010, Southwire Company. All Rights Reserved.
®Southwire is a registered trademark of Southwire Company.